

Provenance-Faithful Physical Reassembly of the TPC Hard Packet

Native-Leaf and Physical Residuals, Exact Block Cover, and Dual Certificates

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Abstract

TPC-117 gives the exact range test for a physical TPC-18 block matrix, while TPC-119 requires every final scalar contribution to remain connected to its original TPC-15 atom. These are not the same test. Starting from a literal native-leaf vector z , let G be the declared leaf-to-physical grouping, C the block-to-leaf synthesis and $B = GC$ the physical block matrix. We compare the canonical residuals

$$r_{\text{leaf}} = (I - CC^\dagger)z, \quad r_{\text{phys}} = (I - BB^\dagger)Gz.$$

They give the two exact decompositions

$$z = CC^\dagger z + r_{\text{leaf}}, \quad Gz = BC^\dagger z + Gr_{\text{leaf}} = BB^\dagger Gz + r_{\text{phys}}.$$

Native-leaf cover implies physical cover, but the converse fails sharply when G hides a provenance direction. Separate dual witnesses therefore stop the two cover claims, and the exact Gram crosswalk is $B^*B = C^*G^*GC$. We also give a coefficientwise test for identifying determinant fibers with zero-mode outer fibers; agreement on one physical vector is insufficient. These are L0 identities and L1 interfaces. The actual growing G, C, z, B archive and its residual estimates remain absent, so the physical reassembly audit is NOT-TESTABLE. No L2 arithmetic saving, endpoint gain or prime-pair theorem is claimed.

1 Why two cover spaces are necessary

TPC-123 replaces the missing global leaf archive by an exact path-expanded object whose columns are original TPC-15 atoms [8, 9]. If that archive is complete, its retained terminal records form a vector

$$z_X = M_X^{\text{ret}} c_X \in \mathbb{C}^{\mathcal{L}^X}.$$

TPC-114 then groups residual labels into native physical labels, whereas TPC-117 asks whether an actual physical vector lies in the range of the TPC-18 block synthesis [3, 4]. Grouping is generally noninjective. It follows that two different statements must be audited:

- (i) every native leaf is synthesized by the declared blocks; and
- (ii) the grouped physical vector is synthesized by the grouped blocks.

The first is a provenance statement relevant to H8. The second is a physical range statement relevant to H6. Neither statement estimates the hard packet.

Definition 1.1 (Claim levels). Finite Hilbert-space identities are L0. Identifying their spaces and maps with the literal TPC records is L1. A uniform growing fixed- h_0 arithmetic power saving is L2.

2 The literal reassembly diagram

Let $\mathcal{L}_X$, $\mathcal{P}_X$, and $\mathcal{Q}_X$ be finite-dimensional Hilbert spaces of native leaves, physical carriers, and block coefficients. The declared maps are

$$C_X : \mathcal{Q}_X \rightarrow \mathcal{L}_X, \quad G_X : \mathcal{L}_X \rightarrow \mathcal{P}_X, \quad B_X = G_X C_X : \mathcal{Q}_X \rightarrow \mathcal{P}_X.$$

Put $z = z_X$, $G = G_X$, $C = C_X$, $B = B_X$, and

$$w = Gz.$$

All adjoints and Moore–Penrose inverses refer to the declared physical Hilbert norms, not to a convenient surrogate norm [1, 2].

Definition 2.1 (Two canonical coefficients and residuals). Define

$$q_{\text{leaf}} = C^\dagger z, \quad r_{\text{leaf}} = (I - CC^\dagger)z, \quad (1)$$

$$q_{\text{phys}} = B^\dagger w, \quad r_{\text{phys}} = (I - BB^\dagger)w. \quad (2)$$

The coefficient vectors need not agree. The first is least squares before grouping; the second is least squares in the physical norm after grouping. Substituting one for the other without proving an isometry is a normalization change.

3 The two-residual theorem

Theorem 3.1 (Provenance–physical reassembly). *The following identities hold exactly:*

$$z = Cq_{\text{leaf}} + r_{\text{leaf}}, \quad (3)$$

$$w = Bq_{\text{leaf}} + Gr_{\text{leaf}}, \quad (4)$$

$$w = Bq_{\text{phys}} + r_{\text{phys}}. \quad (5)$$

Moreover,

$$r_{\text{leaf}} \perp \text{ran}(C), \quad r_{\text{phys}} \perp \text{ran}(B).$$

In particular,

$$r_{\text{leaf}} = 0 \implies r_{\text{phys}} = 0.$$

Proof. The matrices $CC^\dagger$ and $BB^\dagger$ are the orthogonal projections onto $\text{ran}(C)$ and $\text{ran}(B)$, respectively. This proves (3) and (5) together with the orthogonality statements. Applying G to (3) and using $GC = B$ proves (4). If $r_{\text{leaf}} = 0$, then $w = BC^\dagger z \in \text{ran}(B)$, so its orthogonal physical residual vanishes. $\square$

Remark 3.2 (No reverse implication). The theorem does not assert $r_{\text{phys}} = 0 \Rightarrow r_{\text{leaf}} = 0$. A component of r_{leaf} can lie in $\ker(G)$, or its grouped image can fall back into $\text{ran}(B)$. Physical equality alone may therefore erase the missing native route.

Theorem 3.3 (Separate dual certificates). *The two exact cover conditions are*

$$r_{\text{leaf}} = 0 \iff \langle y, z \rangle = 0 \quad \text{for every } y \in \ker(C^*),$$

$$r_{\text{phys}} = 0 \iff \langle v, w \rangle = 0 \quad \text{for every } v \in \ker(B^*).$$

Thus one $y \in \ker(C^*)$ with $\langle y, z \rangle \neq 0$ stops exact native-leaf cover, and one $v \in \ker(B^*)$ with $\langle v, w \rangle \neq 0$ stops exact physical cover.

Proof. For any finite-dimensional operator A , $\text{ran}(A) = \ker(A^*)^\perp$. Apply this to C and B , then use equations (1) and (2). $\square$

4 Scalar evaluation and the native Gram

Let $a \in \mathcal{P}_X$ be the actual physical evaluator and let ν_X be the one inherited normalization.

Corollary 4.1 (Exact scalar return).

$$\boxed{\nu_X \langle a, w \rangle = \nu_X \langle B^* a, q_{\text{phys}} \rangle + \nu_X \langle a, r_{\text{phys}} \rangle.}$$

There is also the provenance-resolving form

$$\nu_X \langle a, w \rangle = \nu_X \langle B^* a, q_{\text{leaf}} \rangle + \nu_X \langle G^* a, r_{\text{leaf}} \rangle.$$

Proof. Pair (5) and (4) with a and use the adjoint identity. $\square$

Proposition 4.2 (Exact native Gram crosswalk). *The physical block Gram is*

$$\boxed{B^* B = C^* G^* G C.}$$

Consequently any determinant, singular value or condition number used after grouping must be computed from the literal G and C . Replacing $G^* G$ by the identity requires an additional isometry theorem.

Proof. Use $B = GC$ and take adjoints. $\square$

This identity attaches the native Gram warning of TPC-114 to the physical range test of TPC-117. It does not say that a large or small condition number is an unavoidable endpoint exponent. TPC-113 and TPC-118 require the actual composite norm and a nonduplicated cost registry [2, 5].

5 The quotient observability operator

The two residuals are not unrelated. Let

$$P_B = BB^\dagger$$

and restrict the operator

$$H = (I - P_B)G|_{\ker(C^*)} : \ker(C^*) \longrightarrow \ker(B^*).$$

This is the part of the physical grouping that remains visible after quotienting out the declared block range.

Theorem 5.1 (Exact residual transfer). *For the residuals in (1)–(2),*

$$\boxed{r_{\text{phys}} = Hr_{\text{leaf}}.} \tag{6}$$

Consequently,

$$\|r_{\text{phys}}\| \leq \|H\| \|r_{\text{leaf}}\| \leq \|G\| \|r_{\text{leaf}}\|.$$

Proof. By (4),

$$w = Bq_{\text{leaf}} + Gr_{\text{leaf}}.$$

Apply $I - P_B$. The first term vanishes because it lies in $\text{ran}(B)$, while the left side becomes r_{phys} . Moreover $r_{\text{leaf}} \in \ker(C^*)$, so the remaining term is exactly Hr_{leaf} . The norm bounds follow from the definition and from $\|I - P_B\| \leq 1$. $\square$

Define the quotient observability constant

$$\gamma_X = \inf_{\substack{r \in \ker(C^*) \\ \|r\|=1}} \|Hr\|,$$

when $\ker(C^*) \neq \{0\}$. In finite dimension, γ_X is the minimum modulus of the restricted operator H on its full declared domain. Equivalently, it is the smallest singular value when zeros coming from $\ker(H)$ are included. Thus $\gamma_X > 0$ precisely when H is injective. The least *positive* singular value on $\ker(H)^\perp$ is a different quantity and does not give a reverse bound for all native residuals when $\ker(H) \neq \{0\}$. If $\ker(C^*) = \{0\}$, then C is already onto the native-leaf space and $r_{\text{leaf}} = 0$ for every z , so no reverse-transfer constant is needed.

Corollary 5.2 (When physical cover detects provenance cover). *If $\ker(C^*) \neq \{0\}$ and $\gamma_X > 0$, then*

$$\gamma_X \|r_{\text{leaf}}\| \leq \|r_{\text{phys}}\| \leq \|H\| \|r_{\text{leaf}}\|,$$

and hence

$$r_{\text{phys}} = 0 \iff r_{\text{leaf}} = 0.$$

The same equivalence holds trivially when $\ker(C^*) = \{0\}$. If $\gamma_X = 0$, there is a nonzero native residual direction that is physically invisible modulo $\text{ran}(B)$.

Proof. When the kernel is zero, $r_{\text{leaf}} = 0$ and [theorem 3.1](#) applies. Otherwise, the lower bound is the definition of γ_X , applied to r_{leaf} . The upper bound is [theorem 5.1](#). In finite dimension, $\gamma_X = 0$ means that H has a nonzero kernel vector. $\square$

Remark 5.3 (A criterion, not a free endpoint gain). A growing lower bound for γ_X could certify that physical residual control transfers back to native provenance control. The present paper neither supplies the actual operator H_X nor proves such a lower bound. Also, γ_X^{-1} would be a norm-conversion cost, not an arithmetic saving; it must be placed in the physical ledger only when the actual proof invokes this reverse transfer.

6 A sharp provenance-camouflage example

Proposition 6.1 (Physical cover can hide a native residual). *There exist G, C, z with $r_{\text{leaf}} \neq 0$ and $r_{\text{phys}} = 0$.*

Proof. In standard Euclidean coordinates take

$$C = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad z = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad G = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Then $r_{\text{leaf}} = z \neq 0$, while $w = Gz = 0$. Hence the physical vector lies in $\text{ran}(GC)$ and $r_{\text{phys}} = 0$. $\square$

Here $H = 0$ on $\ker(C^*) = \mathbb{C}e_3$, so $\gamma_X = 0$. The example is not a claim about the actual TPC archive. It proves that an H6 range certificate cannot be promoted to H8 without either the native-leaf audit or a positive quotient-observability certificate.

7 Determinant fibers and zero-mode fibers

TPC-121 groups raw atoms into determinant bins, while TPC-122 orders literal outer fibers for signed-prefix control [6, 7]. These are different declared dictionaries. Denote their finite index sets by $\mathcal{D}_X$ and $\mathcal{Z}_X$, respectively, and let

$$Q_D : \mathcal{L}_X \rightarrow \mathbb{C}^{\mathcal{D}_X}, \quad Q_Z : \mathcal{L}_X \rightarrow \mathbb{C}^{\mathcal{Z}_X}$$

be their exact leaf-level aggregation matrices. A proposed identification $J : \mathbb{C}^{\mathcal{D}_X} \rightarrow \mathbb{C}^{\mathcal{Z}_X}$ must be proved, not inferred from similar symbols. The scalar determinant energy D_X and scalar distinguished zero mode Z_X retain their earlier meanings and are not index sets.

Theorem 7.1 (Coefficientwise fiber-intertwining test). *Let $M : \mathbb{C}^{\mathcal{A}_X} \rightarrow \mathcal{L}_X$ be the complete native-leaf matrix. The two fiber descriptions agree on every original packet coefficient vector precisely when*

$$(JQ_D - Q_Z)M = 0.$$

Agreement on one coefficient vector c_X , $(JQ_D - Q_Z)Mc_X = 0$, is not sufficient for this matrix identity.

Proof. The two descriptions agree on every input if and only if the two linear maps JQ_DM and Q_ZM are equal, which is the displayed matrix condition. Equality after applying them to one vector tests only one element of their difference kernel. $\square$

Remark 7.2 (No exponent transfer). Even if [theorem 7.1](#) passes, TPC-121 and TPC-122 provide only conditional formulas for λ_D^{cert} and η_Z^{cert} . They do not prove their growing inputs, and their determinant reserve cannot be charged against the physical endpoint ledger.

8 Typed invocation and route decisions

Definition 8.1 (Paper-specific verdicts). GO-INTERFACE means that the indicated exact map and all its quantifiers are present. STOP-DECLARED-ROUTE means that an exact dual witness, normalization mismatch or provenance defect stops the declared reassembly. It does not stop every possible representation. NOT-TESTABLE means that the actual growing matrix or typed vector required to execute a test is absent.

Target	GO-INTERFACE input	Stop or incomplete outcome
Native-leaf cover	Complete z, C with $r_{\text{leaf}} = 0$	Leaf dual witness gives STOP-DECLARED-ROUTE; missing z, C gives NOT-TESTABLE.
Physical exact cover	Complete w, B with $r_{\text{phys}} = 0$	Physical dual witness gives STOP-DECLARED-ROUTE; missing w, B gives NOT-TESTABLE.
Physical residual route	Same normalization and a theorem $\nu_X \langle a, r_{\text{phys}} \rangle = o(X)$	Nonzero residual alone stops exact cover, not the residual route.
H8 reconnection	TPC-123 path conservation plus native-leaf cover and complete soft bound	Physical cover alone is insufficient.
H9 endpoint	Actual growing norms and a complete TPC-118 registry	Unknown costs remain unknown, never zero.

At present, TPC-123 supplies the schema rather than the actual growing leaf vector, and TPC-117 reports that the literal growing B, w archive is unavailable. Hence

$$\text{Reassembly}_{124} = \text{NOT-TESTABLE from current artifacts.}$$

This is an executable-data verdict, not an impossibility theorem.

9 Reproducible audit and claim boundary

The standard-library script `experiments/tpc124_reassembly_audit.py` uses exact rational coordinate models. It verifies both residual decompositions, a dual witness, the scalar return, [proposition 4.2](#), and both extremal quotient-observability regimes. It also constructs [proposition 6.1](#) and two distinct fiber dictionaries that agree on one vector but not coefficientwise. The model is not the missing physical archive.

Statement	Status
Two-residual and dual theorems	Exact L0.
Native Gram and scalar crosswalk	Exact L0; literal attachment is L1.
Quotient observability criterion	Exact L0; no growing γ_X bound.
Determinant/zero-fiber comparison rule	Exact L0/L1 interface.
Actual growing G, C, z, B	Absent; NOT-TESTABLE.
Complete physical residual $o(X)$	Not proved.
Growing fixed- h_0 arithmetic saving	Not proved; no L2 event.

Explicit exclusions. This paper proves no complete physical cover for the TPC packet, no H8 conjunction, no frame or localization exponent, no endpoint below $1/400$, no determinant-energy or zero-mode exponent, no signed Möbius cancellation, no parity breakthrough, no Hardy–Littlewood asymptotic, and no prime-pair or twin-prime theorem.

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